

CUSTOMER CHURN PREDICTION

HELPING SYRIATEL KEEP MORE CUSTOMERS

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Phase 3 Project

Overview

- SyriaTel is losing customers every month.
- Losing customers means losing revenue.
- It is much cheaper to keep current customers than to find new ones.
- The goal is to create a simple tool that predicts which customers are likely to leave soon.

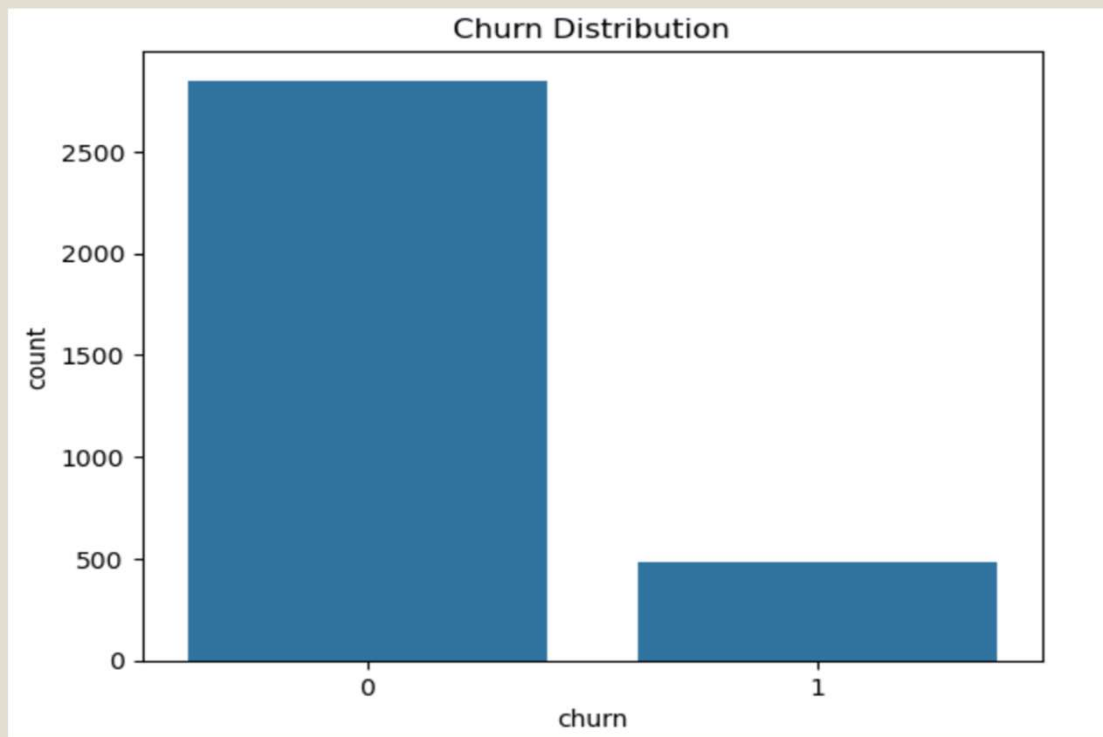
Business Problem

- Customer churn is a major challenge for SyriaTel.
- The company needs to identify customers likely to leave.
- Early intervention can reduce customer loss.
- The project aims to support customer retention decisions using machine learning.
- **Stakeholders:**
 - ❖ Customer Retention Team
 - ❖ Marketing Team
 - ❖ Customer Support Department

Data Understanding

- Dataset contains information from 3,333 telecom customers.
- **Features include:**
 - ❖ call usage
 - ❖ international plan
 - ❖ voicemail plan
 - ❖ customer service calls
 - ❖ customer churn status.
- **Dataset contained:** No missing values and No duplicate records.
- **Key Observation:** Only 14.5% of customers churned.

Data Understanding: Churn Distribution



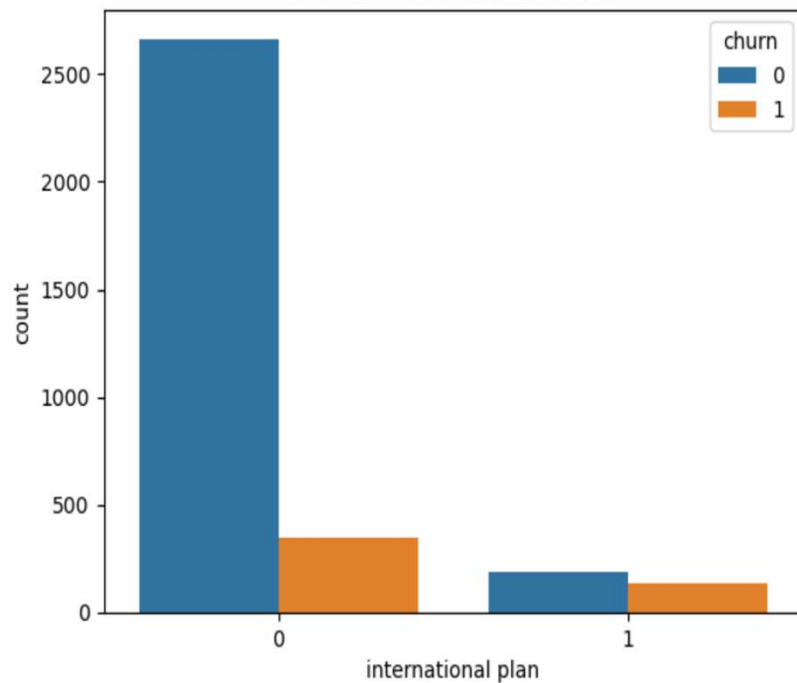
Exploratory Data Analysis

- **Key Behavioural Patterns Identified:**

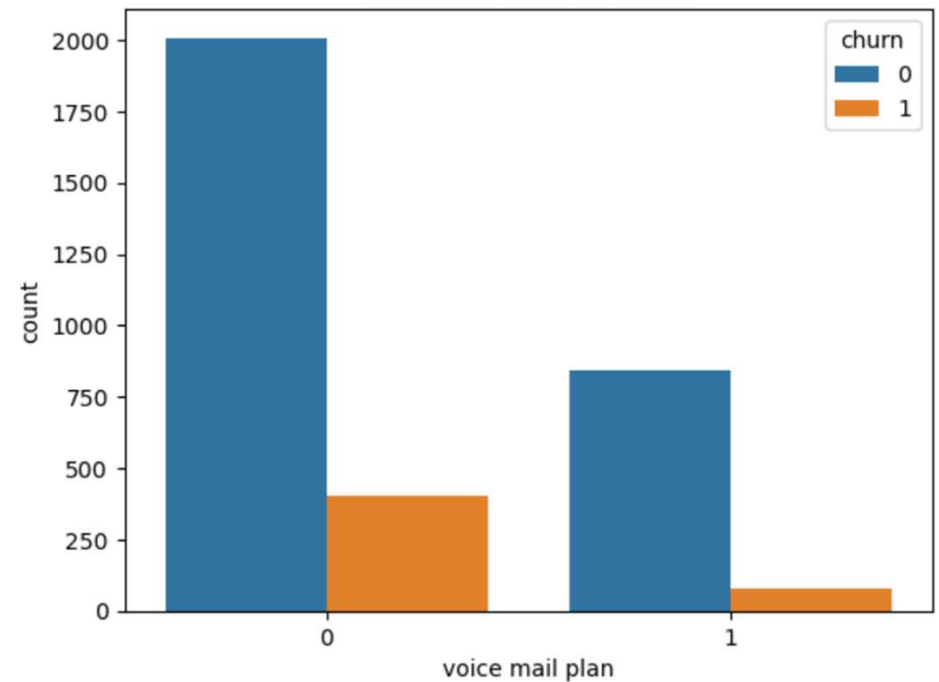
- ❖ Customers with international plans churned more frequently.
- ❖ Customers with many customer service calls were more likely to churn.
- ❖ Customers with voicemail plans were less likely to churn.

Exploratory Data Analysis

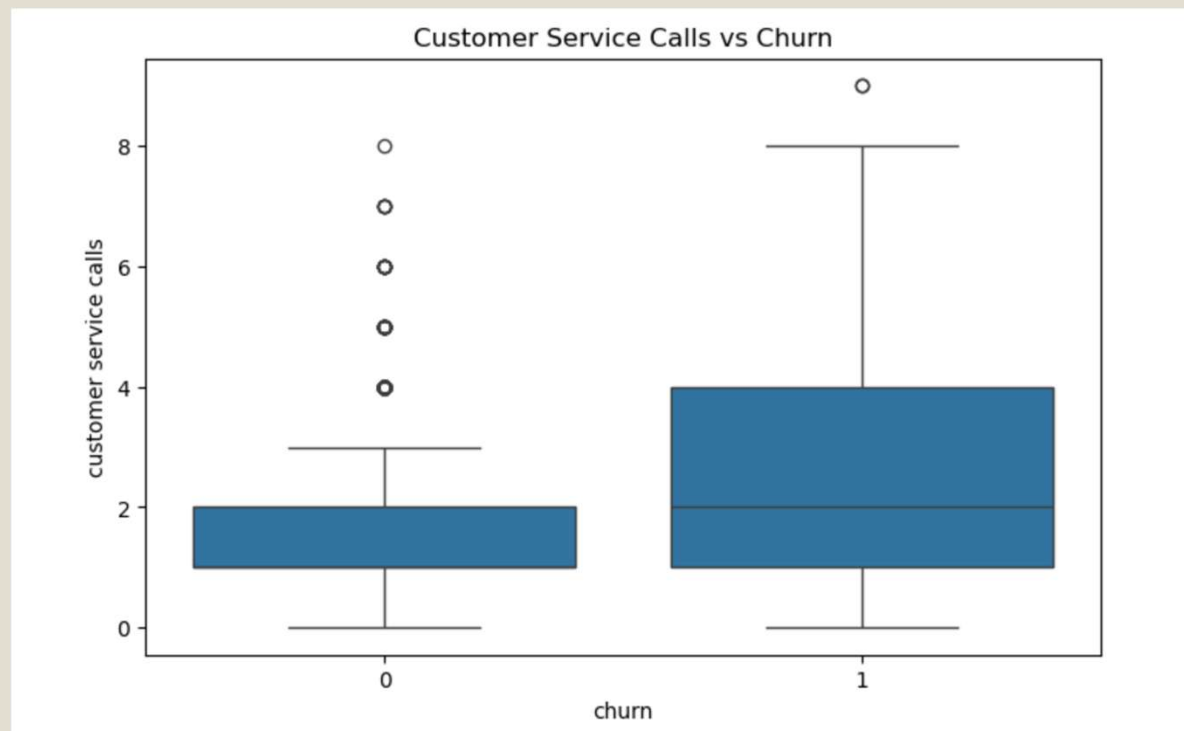
International Plan vs Churn



Voice Mail Plan vs Churn



Exploratory Data Analysis



Data Preparation

- **Data Cleaning and Transformation:**
- Removed irrelevant columns:
 - ❖ Phone number
 - ❖ State
 - ❖ Area code
- Removed highly correlated charge variables.
- Encoded categorical variables into numerical format.
- Scaled features for Logistic Regression modeling.

Modelling Approach

- **Why Classification?**

- ❖ Classification helps predict whether a customer will Stay or Churn

- **Models Tested:**

- ❖ Baseline Logistic Regression

- ❖ Improved Logistic Regression

- ❖ Decision Tree

- ❖ Tuned Decision Tree

- The goal is to Identify the model that best predicts churn customers.

Logistic Regression Results

- **Baseline Logistic Regression:**
 - ❖ High overall accuracy (85%)
 - ❖ Poor churn detection recall (17%)
 - ❖ Missed many actual churn customers
- **Improved Logistic Regression:**
 - ❖ Recall improved to 77%
 - ❖ Better at identifying churn customers
 - ❖ Accuracy reduced slightly

Logistic Regression Comparison

Accuracy Score:
0.8545727136431784

Classification Report:

	precision	recall	f1-score	support
0	0.87	0.98	0.92	566
1	0.57	0.17	0.26	101
accuracy			0.85	667
macro avg	0.72	0.57	0.59	667
weighted avg	0.82	0.85	0.82	667

Confusion Matrix:

```
[[553 13]
 [ 84 17]]
```

Accuracy Score:
0.7841079460269865

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.79	0.86	566
1	0.39	0.77	0.52	101
accuracy			0.78	667
macro avg	0.67	0.78	0.69	667
weighted avg	0.87	0.78	0.81	667

Confusion Matrix:

```
[[445 121]
 [ 23 78]]
```

Decision Tree Results

- **Decision Tree Performance:**

- ❖ Accuracy: 94%

- ❖ Precision: 88%

- ❖ Recall: 67%

- ❖ F1-Score: 76%

- **Why It Performed Better:**

- ❖ Captured complex customer behavior patterns

- ❖ Balanced precision and recall effectively

Decision Tree Results

Accuracy Score:

0.9370314842578711

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.98	0.96	566
1	0.88	0.67	0.76	101
accuracy			0.94	667
macro avg	0.91	0.83	0.86	667
weighted avg	0.93	0.94	0.93	667

Confusion Matrix:

```
[[557  9]
 [ 33 68]]
```

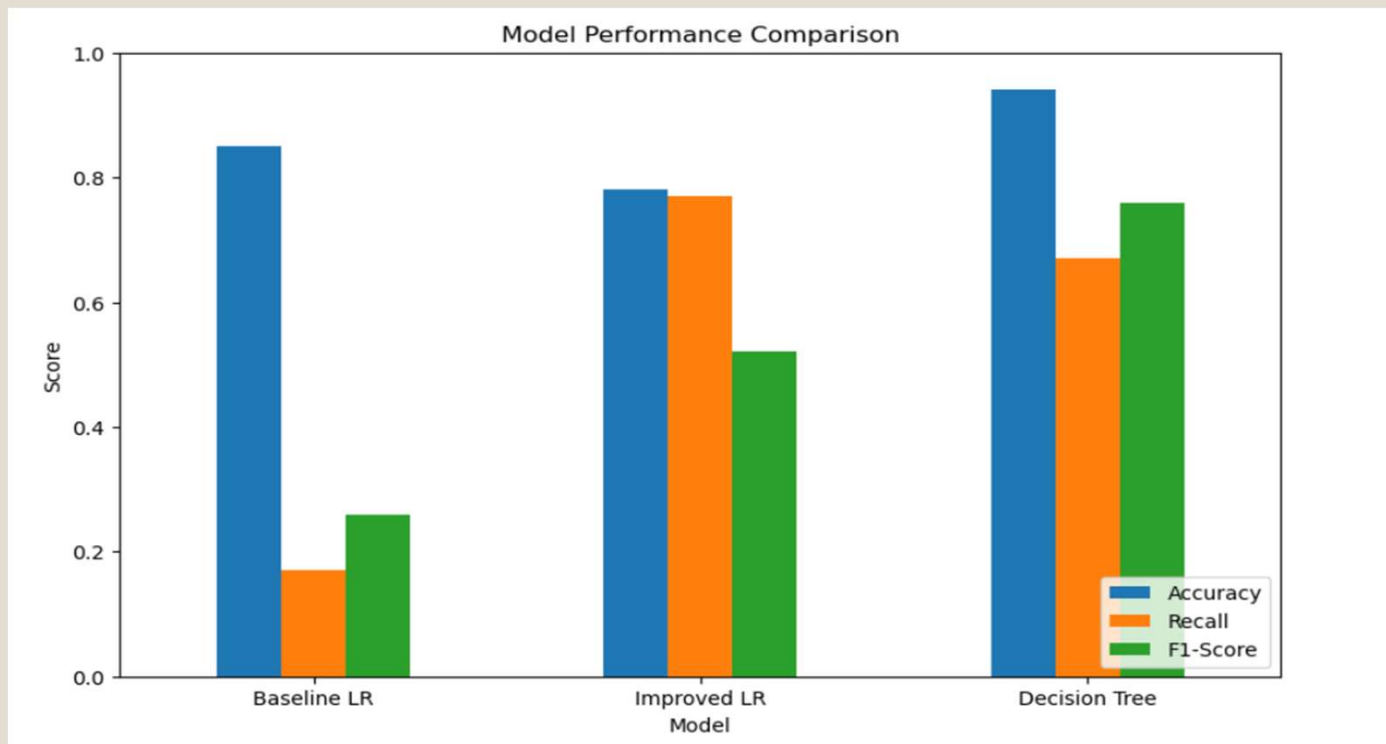
Model Comparison

	Model	Accuracy	Precision (Churn)	Recall (Churn)	F1-Score (Churn)
0	Baseline Logistic Regression	0.854573	0.57	0.17	0.26
1	Improved Logistic Regression	0.784108	0.39	0.77	0.52
2	Decision Tree	0.937031	0.88	0.67	0.76
3	Tuned Decision Tree	0.935532	0.88	0.66	0.76

Conclusion:

- Decision Tree achieved the best balance.
- Selected as final model.

Model Performance Comparison



Key Insights

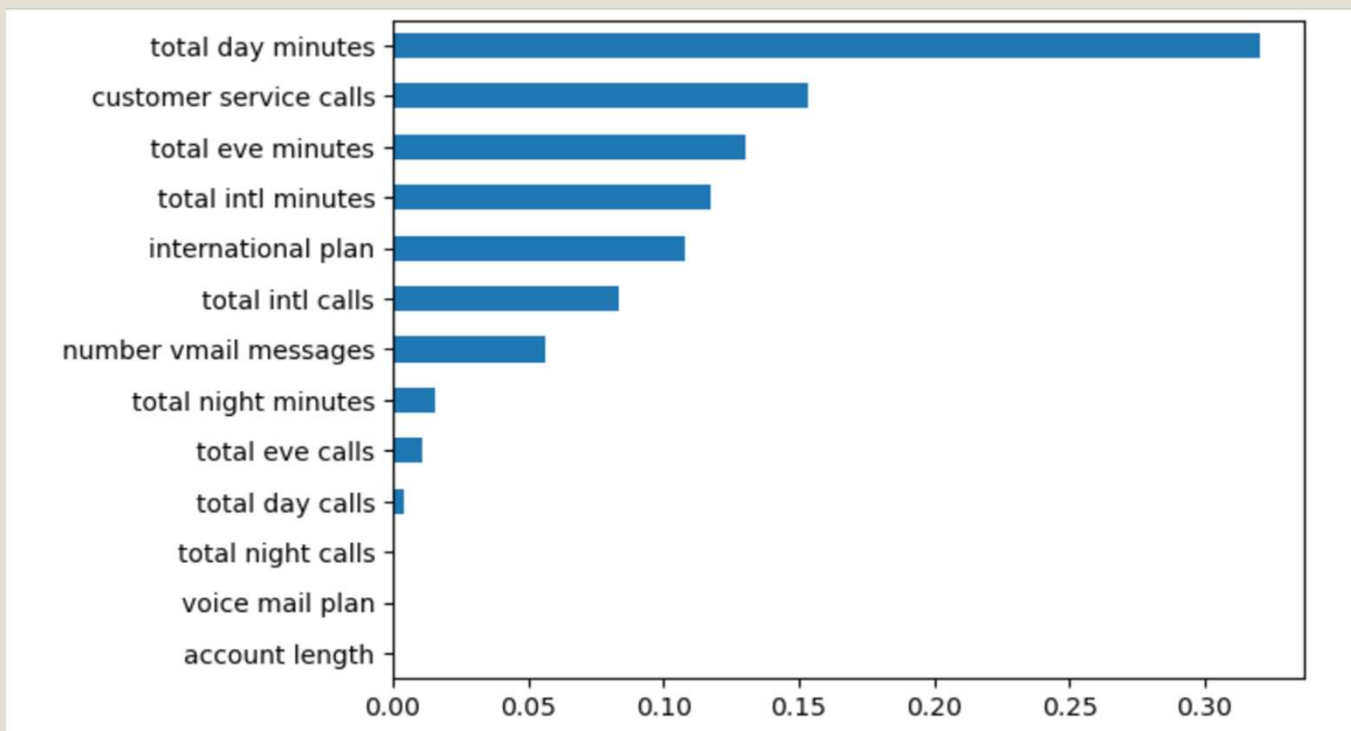
- **Important Churn Drivers**

- ❖ High customer service calls increased churn risk.
- ❖ International plan customers churned more .
- ❖ Voice mail plan customers tended to stay longer .

- **Business Meaning**

- ❖ Customer behavior patterns can help SyriaTel intervene before customers leave.

Key Insights



Business Recommendations

- **Recommended Actions:**

- ❖ Improve customer service quality
- ❖ Review international plan pricing
- ❖ Strengthen retention campaigns
- ❖ Use predictive analytics monthly
- ❖ Increase customer engagement initiatives

Next Steps

- Deploy model into operational systems.
- Continuously retrain with updated data.
- Monitor churn prediction performance.
- Combine predictions with retention offers.

Thank You!